Python Coding Assignment - Intern 1

Modules 1 and 2 | Theme: Retail Sales | 15 Questions

Name: _____ Date: _____

Instructions

1. Write and run every program in VS Code. Save each one as a separate file: q1.py, q2.py ... q15.py.
2. Your output must match the Expected Output. For programs that use input(), your prompt text can be different, but the result lines must be the same.
3. Use only the topics covered in Module 1 and Module 2 of the course.
4. Add a comment with your name at the top of every file.
5. Put all files in one folder named Intern1_Assignment and submit it.

Q1. Receipt Printer *[Formatted Output]*

Create variables item = "Keyboard", price = 1499.5 and qty = 3. Using an f-string, print one line in exactly this format: Item: Keyboard | Price: Rs. 1499.50 | Qty: 3 | Total: Rs. 4498.50 (price and total must show 2 decimal places).

Expected Output

```
Item: Keyboard | Price: Rs. 1499.50 | Qty: 3 | Total: Rs. 4498.50
```

Q2. Profit Calculator *[Variables, Input, Arithmetic]*

Take the cost price and the selling price from the user. Print the profit and the profit percentage, both with 2 decimal places.

Sample Input (one value per line)

```
800
1000
```

Expected Output

```
Enter cost price: 800
Enter selling price: 1000
Profit: 200.00
Profit %: 25.00
```

Q3. Order Total from Text *[Type Conversion]*

The user enters the quantity and the unit price. Convert them to the right types (int for quantity, float for price). Print the total amount, and then the total as a whole number using int().

Sample Input (one value per line)

```
12
49.5
```

Expected Output

```
Quantity: 12
Unit price: 49.5
Total: 594.0
Whole rupees: 594
```

Q4. Target Check *[Comparison and Logical Operators]*

Take the monthly sales and the sales target from the user. Print three lines: Target achieved (sales are at least the target), Double target (sales are at least twice the target) and Bonus eligible (target achieved AND sales are at least 50000). Each must be True or False.

Sample Input (one value per line)

```
62000
55000
```

Expected Output

```
Sales: 62000
Target: 55000
```

```
Target achieved: True
Double target: False
Bonus eligible: True
```

Q5. GST with Comments *[Comments]*

Take an amount from the user and add 18% GST. Print the GST and the final amount (2 decimal places). Your program must start with a comment that has your name and today's date, and every step must have its own comment explaining it (at least 4 comments in total).

Sample Input (one value per line)

```
2500
```

Expected Output

```
Enter amount: 2500
GST: 450.00
Final amount: 2950.00
```

Q6. Monthly Sales Summary *[Lists]*

monthly_sales = [12000, 15000, 9800, 17500, 14200, 16800]. Print the total sales, the average (2 decimal places), the highest, the lowest and the number of months.

Expected Output

```
Total: 85300
Average: 14216.67
Highest: 17500
Lowest: 9800
Months: 6
```

Q7. Product Tuple *[Tuples]*

product = ("Laptop", 55000, "Electronics"). Unpack it into name, price and category and print: Laptop belongs to Electronics and costs Rs. 55000. Then, for quarterly = (10, 20, 30, 20, 40, 20), print how many times 20 appears, the index of 30 and the length of the tuple.

Expected Output

```
Laptop belongs to Electronics and costs Rs. 55000
Count of 20: 3
Index of 30: 2
Length: 6
```

Q8. Customer Overlap *[Sets]*

jan = {"Amit", "Sara", "John", "Riya"} and feb = {"Sara", "Riya", "Kunal", "Neha"}. Print (as sorted lists, using sorted()): customers in both months, customers in either month, customers only in January, customers only in February. Then print the number of unique customers.

Expected Output

```
Both: ['Riya', 'Sara']
Either: ['Amit', 'John', 'Kunal', 'Neha', 'Riya', 'Sara']
Only Jan: ['Amit', 'John']
Only Feb: ['Kunal', 'Neha']
Unique customers: 6
```

Q9. Stock Dictionary *[Dictionaries]*

inventory = {"pen": 50, "book": 20, "bag": 10}. Add "pencil" with 100 units, change the stock of "book" to 35, and remove "bag" using pop(). Print the final dictionary and the total stock using sum() on the values.

Expected Output

```
{'pen': 50, 'book': 35, 'pencil': 100}
Total stock: 185
```

Q10. Month Slicer *[Indexing and Slicing]*

months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]. Print: the first month, the last month (use a negative index), the first quarter (Jan to Mar), the last 3 months, every second month starting from Jan, and the whole list in reverse order (use slicing).

Expected Output

```
Jan
Dec
['Jan', 'Feb', 'Mar']
['Oct', 'Nov', 'Dec']
['Jan', 'Mar', 'May', 'Jul', 'Sep', 'Nov']
['Dec', 'Nov', 'Oct', 'Sep', 'Aug', 'Jul', 'Jun', 'May', 'Apr', 'Mar', 'Feb', 'Jan']
```

Q11. Shopping Cart *[List Operations]*

Start with cart = ["pen", "book"]. In this order: append "bag", insert "ruler" at index 0, extend with ["eraser", "pencil"], remove "book". Print the list, then sort it and print it with its length, then reverse it and print it again.

Expected Output

```
['ruler', 'pen', 'bag', 'eraser', 'pencil']
['bag', 'eraser', 'pen', 'pencil', 'ruler'] 5
['ruler', 'pencil', 'pen', 'eraser', 'bag']
```

Q12. Sales Table *[Nested Lists]*

table = [["Jan", 1200], ["Feb", 1500], ["Mar", 900]]. Print the sales of Feb using two indexes. Add the row ["Apr", 1700]. Using a for loop, print each row as Month: Jan, Sales: 1200 and then print the total sales.

Expected Output

```
1500
Month: Jan, Sales: 1200
Month: Feb, Sales: 1500
Month: Mar, Sales: 900
Month: Apr, Sales: 1700
Total: 5300
```

Q13. Store Records *[Nested Dictionaries]*

stores = {"Pune": {"sales": 50000, "staff": 8}, "Mumbai": {"sales": 82000, "staff": 12}, "Nashik": {"sales": 31000, "staff": 5}}. Print the sales of Mumbai. Change the staff of Nashik to 6. Add "Nagpur" with sales 27000 and staff 4. Using a loop, print every store as Pune: sales=50000, staff=8 .

Expected Output

```
82000
Pune: sales=50000, staff=8
Mumbai: sales=82000, staff=12
Nashik: sales=31000, staff=6
Nagpur: sales=27000, staff=4
```

Q14. Festival Discount *[Iterating]*

prices = [499, 1299, 250, 3999, 150]. Using a for loop, print each price with a 10% discount in the format Original: 499 | Discounted: 449.10 (2 decimal places). After the loop, print the total saving.

Expected Output

```
Original: 499 | Discounted: 449.10
Original: 1299 | Discounted: 1169.10
Original: 250 | Discounted: 225.00
Original: 3999 | Discounted: 3599.10
Original: 150 | Discounted: 135.00
Total saving: 619.70
```

Q15. Product Stock Report *[Recap (all data structures)]*

products = {"Mouse": [499, 10], "Keyboard": [1299, 5], "Monitor": [8999, 3], "Cable": [199, 40]} (each value is [price, quantity]). Loop through the dictionary and print Mouse: stock value = 4990 for each product. Then print the grand total stock value, the number of products and the product names as a sorted list.

Expected Output

```
Mouse: stock value = 4990
Keyboard: stock value = 6495
Monitor: stock value = 26997
Cable: stock value = 7960
Grand total: 46442
Products: 4
Names: ['Cable', 'Keyboard', 'Monitor', 'Mouse']
```